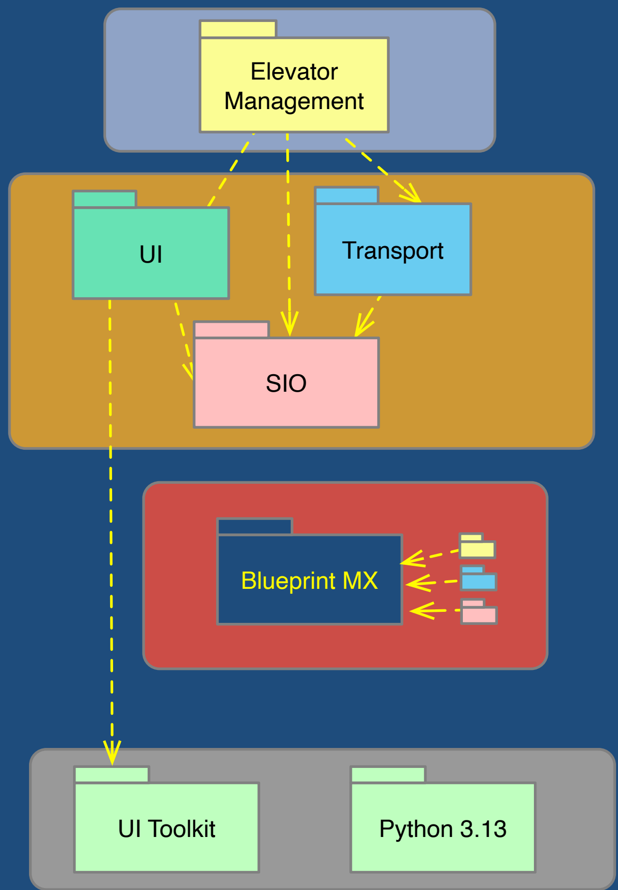
Domain Diagram

Conceptual Layers



Application

The business purpose of the system (Always a single domain at this layer)

Modeled elements (classes, state models, etc.) are highly specific to the application.

This is where the fundamental business-system logic is defined.

Service

Each domain at this layer provides services that could conceivably be useful to an entirely different application.

Modeled elements are generic relative to the application

But the data populated into these domains IS application specific

Model Execution (MX)

An essential service for all modeled domains since this is what runs the models.

Here the language and its execution logic is modeled. (The Shlaer-Mellor Metamodel)

While the metamodel does not vary across software platforms, the MX implementation is specific to a particular category of platform (embedded, distributed, multi-threaded)

Our book Models to Code for a detailed explanation of these categories and implementation strategies.

Software Platform

The Blueprint MX implemented in Python 3.13.

The UI is realized via SIO where blinking lights and physical buttons are concerned and in some kind of UI toolkit for the administrative interface.

A variety of MX's have been used over the years to implement versions of the Elevator Case Study system. A real world elevator system, of course, would be built on an entirely different software platform.

Nonetheless, the domains in the Application and Service layers should not be affected if modeled properly.

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Domain Diagram **Elevator 3 Case Study**	Leon Starr Aug 9, 2026
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